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ENERGY

# Seconds of Spin — Inertia and the Physics of Keeping a Grid Together

*Every power system runs on a buffer of kinetic energy measured in seconds, and the whole architecture of grid reliability is an argument about how many seconds are enough.*

2,705 words · 14-minute read

## A Lightning Strike in Hertfordshire

Lightning hit a 400 kV transmission circuit north of London at 16:52:33.490 on 9 August 2019. Protection cleared the fault in seventy milliseconds at one end and seventy-four at the other, which is roughly a quarter of the time it takes to blink. By any conventional reckoning the grid had done exactly what it was built to do. Nine seconds of ordinary weather should have been the end of it.

Instead, within two seconds Great Britain had lost 981 megawatts of generation: the Hornsea offshore wind farm reduced output by 737 MW and the steam turbine at Little Barford tripped, taking 244 MW with it. Seventy-seven seconds after the lightning, at a system frequency of 48.8 Hz, an automatic scheme opened circuit breakers on 931 MW of demand — 3.2 per cent of the country — and about 1.15 million customers went dark [1].

The system had been secured that afternoon against the loss of its single largest infeed, 1,000 MW [1]. The standard was not obviously wrong; the initial loss was inside it. What decided the outcome was a quantity that almost never appears in public argument about electricity: how much mass was spinning in Britain at that moment, and how fast. The system operator put the figure at roughly 210 GVA·s [1], a number the academic post-mortem refined to about 220 GVA·s falling to 212 GVA·s during the event [2] — agreement close enough that nothing below turns on the difference.

Convert it into energy and it becomes startling. Two hundred and ten gigawatt-seconds is 210 gigajoules, or 58 megawatt-hours. The entire mechanical buffer standing between a lightning strike and a million dark houses held less than half the energy of the Hornsdale Power Reserve in South Australia as originally built, a single battery installation rated 100 MW and 129 MWh [3]. Grids are not, it turns out, massive things. They are fast things with almost no cushion.

## The Ledger That Cannot Be Overdrawn

Electricity is the rare commodity with no inventory. Generation and consumption must match continuously, and the matching is not enforced by a controller — it is enforced by Newton. Every synchronous generator on an alternating-current network turns in electrical lockstep with every other, so the network's frequency is a single shared tachometer reading of a continent's worth of rotating steel. When demand exceeds generation, the shortfall is not politely queued. It is taken out of the rotors as mechanical work, and they slow down.

The bookkeeping is the swing equation, the foundational relation of power-system dynamics [4]. For a machine of rated apparent power  $S_B$  carrying mechanical input  $P_{\text{mech}}$  and electrical output  $P_{\text{elec}}$ ,

$$\frac{2H}{f_0} \frac{df}{dt} = \frac{P_{\text{mech}} - P_{\text{elec}}}{S_B}$$

where  $f_0$  is nominal frequency and  $H$  is the inertia constant, defined as the stored kinetic energy normalised by rating,

$$H = \frac{E_{\text{kin}}}{S_B} = \frac{J \Omega_r^2}{2 S_B}.$$

That definition rewards a moment's attention, because  $H$  has units of seconds and a physical meaning attached to them: it is how long the machine could supply its own rated power using nothing but the kinetic energy already in its rotor. For conventional turbogenerators  $H$  falls in the range of two to ten seconds [5], and the aggregate figure for a whole synchronous area behaves the same way. Analysis of the German system found an aggregate inertia constant fluctuating between a nominal six seconds and periods of three to four seconds, with inertia below four seconds for 293 hours — some 3.3 per cent of the year — as early as 2012 [5].

The hardware behind those seconds is worth picturing. A 500 MVA turbogenerator with  $H = 4$  s stores 2 gigajoules. At 3,000 rpm, or 314 rad/s, that implies a moment of inertia near 40,000 kg·m<sup>2</sup>: a rotor line tens of metres long, weighing on the order of a hundred tonnes, turning fifty times a second for years without stopping. This is not a metaphor for stability. It is the stability, in steel.

## How Fast the Needle Falls

The quantity operators actually watch is not the frequency but its derivative. Take the swing equation, aggregate it over a synchronous area, and evaluate it the instant after a power imbalance appears and before any control system has responded. What emerges is the rate of change of frequency, or RoCoF [6]:

$$\left. \frac{df}{dt} \right|_{t=0+} = \frac{\Delta P}{P_{\text{load}}} \cdot \frac{f_0}{2H} = \frac{\Delta P f_0}{2 E_{\text{kin}}}.$$

The second form is the useful one. The initial slope is set by two numbers only: the size of the imbalance and the total kinetic energy on the system. Nothing about the technology of the lost plant enters, and nothing about reserves — those act later.

Return to Britain on that August afternoon. An imbalance of 981 MW against 210,000 MW·s of stored energy at 50 Hz gives an initial RoCoF of 0.117 Hz/s. Now note the number that turns out to matter: distributed generators embedded in the British distribution networks carried loss-of-mains protection set to trip at 0.125 Hz/s [1], [2]. The first loss came in just under the threshold, with almost nothing to spare.

It did not stay there. A separate and older protection type, vector-shift relays, disconnected about 150 MW of embedded generation almost immediately on the fault itself. That pushed the imbalance to 1,131 MW, and the same arithmetic now returns 0.135 Hz/s — above the threshold. Several hundred megawatts of additional distributed generation duly tripped on RoCoF, deepening the imbalance, and the frequency continued down to the demand-disconnection setting [1], [2]. The published analysis reaches the same conclusion from full dynamic simulation: eliminate that initial 150 MW vector-shift loss and the RoCoF threshold is never breached, roughly 480 MW of further generation stays connected, and the load-shedding scheme is never armed [2]. A cascade that cost a million customers their supply turned on a margin of 0.010 Hz/s.

### CONSIDER

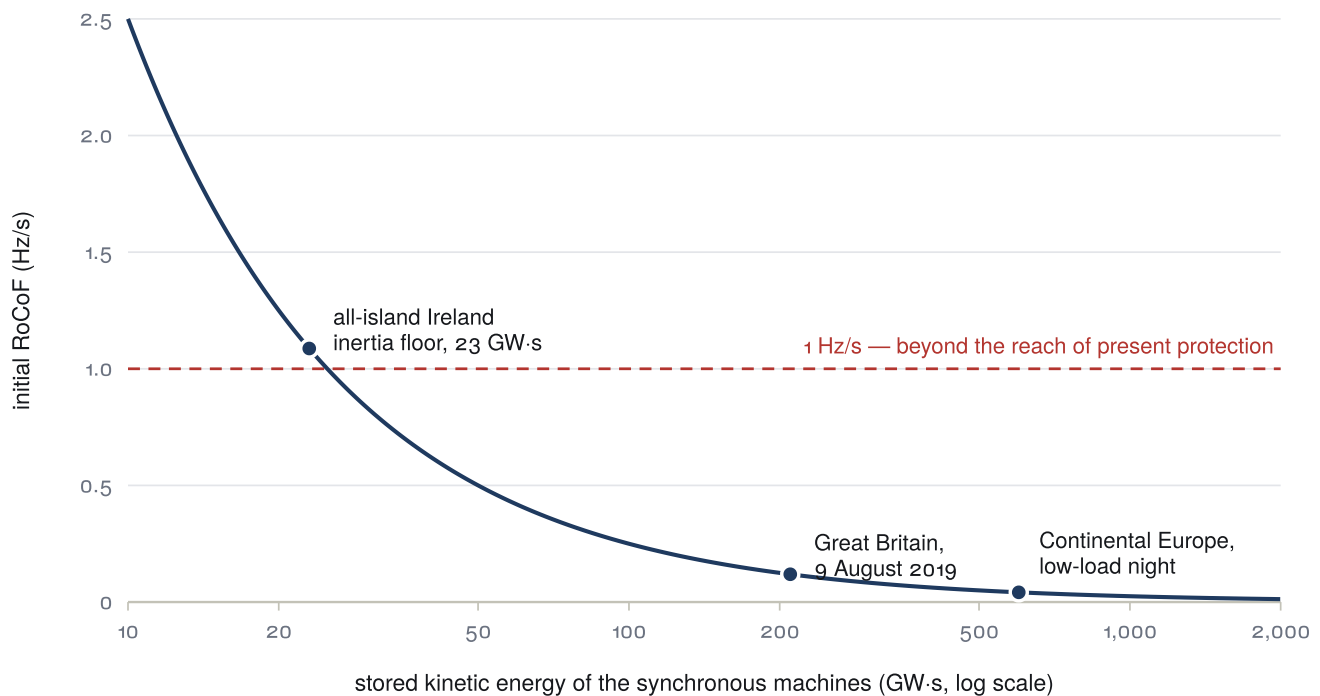
Suppose the same 981 MW had been lost with only half as much kinetic energy on the system. The initial slope doubles, to about 0.23 Hz/s. Does the frequency nadir also double in depth?

It does not, and the reason is that the nadir is not a property of inertia at all. Inertia fixes the opening slope; the depth of the dip is set by a race between that slope and the arrival of governor and fast-frequency response over the following seconds. Halving inertia gives those controllers half as long to act, so the nadir does deepen — but by how much depends entirely on how fast and how large the response is. On 9 August the frequency was arrested at 49.1 Hz about twenty-five

seconds in, then fell again only when a second Little Barford unit tripped a minute later [1]. Counterfactual simulation found that full delivery of the response that had actually been procured — 1,338 MW of it — would have held the frequency above 49.0 Hz throughout [2]. The inertia was never the binding constraint. It was the clock.

## The Price of a Second

Because inertia sets a clock rather than a quantity, system operators do not buy it the way they buy energy. They impose floors on it, and then constrain everything else to fit underneath.



The all-island Irish system, small and weakly interconnected, states its constraints explicitly. Operational policy has held system inertia above 23,000 MW·s, capped instantaneous non-synchronous penetration at 70 per cent of demand, and limited RoCoF to 0.5 Hz/s — with a 75 per cent penetration trial begun in April 2021 and a 1 Hz/s RoCoF trial in June 2020 [7]. Those numbers are not independent of one another.

### CONSIDER

At an inertia floor of 23,000 MW·s and a RoCoF limit of 0.5 Hz/s on a 50 Hz system, what is the largest instantaneous loss the rules will tolerate?

Rearranging the RoCoF expression gives  $\Delta P = 2E_{\text{kin}} \cdot (df/dt)/f_0$ , or 460 MW — and 920 MW if the 1 Hz/s trial becomes policy. The inertia floor and the RoCoF limit together are a statement about how large a single generator or interconnector Ireland is willing to have on its system at all. Loosening the RoCoF limit is therefore not a safety concession for its own sake; it is what buys room for the next large connection.

Texas runs the same calculation at a different scale. ERCOT identifies a “critical inertia” level of 100 GW·s, below which it cannot ride through its design contingency: simultaneous loss of two units at the South Texas Project, 2,750 MW [8]. At 60 Hz that combination implies an initial RoCoF of 0.83 Hz/s, leaving about 0.85 seconds before the first stage of under-frequency load shedding at 59.3 Hz. ERCOT’s own dynamic simulations, condensed into a regression on inertia, give 0.465

seconds for the frequency to travel from 59.7 Hz to 59.3 Hz at that inertia level; the linear estimate above gives 0.485 seconds for the same interval — a four per cent discrepancy, which is about what one should expect between a straight-line extrapolation of an initial slope and a full simulation that includes load damping and early governor action [8].

Actual Texan inertia sits well above the floor, typically between 200 and 400 GW·s, with a record low of 115 GW·s in March 2022 [9]. That margin shapes policy. In September 2024, having weighed rising minimum load, incoming synchronous condensers and grid-forming inverters, ERCOT concluded that a dedicated inertia product was not needed [9].

Set the four cases side by side and the hyperbola in the figure acquires a policy meaning. The same tolerable slope is reached at radically different scales, because what matters is the ratio of the credible loss to the stored energy, not either quantity on its own.

| System                           | Kinetic energy | Reference loss | Initial RoCoF |
|----------------------------------|----------------|----------------|---------------|
| All-island Ireland, policy floor | 23,000 MW·s    | 460 MW         | 0.50 Hz/s     |
| Great Britain, 9 August 2019     | 210,000 MW·s   | 981 MW         | 0.12 Hz/s     |
| ERCOT, critical inertia level    | 100,000 MW·s   | 2,750 MW       | 0.83 Hz/s     |
| Continental Europe, low load     | 600,000 MW·s   | 3,000 MW       | 0.13 Hz/s     |

Ireland and Continental Europe sit at almost the same RoCoF while differing by a factor of twenty-six in stored energy, and Ireland reaches it with a loss barely a seventh the size. A small system is not fragile because its machines are worse. It is fragile because a single unit is a large fraction of the whole — and there is no way to engineer that ratio away except by limiting how large any one unit may be, which is precisely what an inertia floor paired with a RoCoF limit does.

Great Britain reached the opposite conclusion and went shopping. Its stability procurement contracted 6.75 GVA·s of inertia and 11.55 GVA of short-circuit level in Scotland for about 323 million pounds, against an estimated 500 million pounds of avoided cost over ten years — and the second phase brought the first grid-forming batteries online in Scotland in March 2025 [10]. Two competent operators, the same physics, opposite procurement decisions. The difference is not disagreement about the swing equation. It is that Texas is large and synchronously isolated while Britain is small, wind-rich and stitched to the continent by direct-current cables that contribute no inertia at all.

## Two Ways to Lose a Grid

The canonical inertia-and-frequency disaster remains the Continental European system split of 4 November 2006. A 380 kV line was switched off over the Ems river in northern Germany to let a ship pass; loading redistributed, and at 22:10:28 the synchronous area tore into three. The western area found itself short roughly 8,940 MW of imports and fell to about 49 Hz, while the north-eastern fragment, now long on generation, rose to about 51.4 Hz. Automatic schemes shed some 17,000 MW of demand — nearly twice the deficit, an overshoot that is itself instructive — and more than fifteen million households lost supply. Resynchronisation took 38 minutes [11]. Continental Europe is enormous, which is why 49 Hz was as bad as it got: against a reference incident of 3,000 MW, its ordinary operating band is 49.8 to 50.2 Hz and blackout becomes unavoidable only beyond 47.5 Hz [12].

South Australia on 28 September 2016 shows the other failure mode, where protection settings rather than raw inertia do the damage. Tornadoes took out transmission towers; nine wind farms, hitting a pre-set limit on the number of voltage disturbances they would ride through, cut output by 456 MW in under seven seconds. That transferred load onto the Heywood interconnector to Victoria, whose protection scheme opened at 16:18:15.8, and the state was black about a second later. Under-frequency load shedding never had a chance: it was armed at 49 Hz, and the rate of frequency change after

separation was simply too fast for it to arrest the fall [13]. ENTSO-E's assessment of the general case is blunt — RoCoF above roughly 1 Hz/s is not survivable with present protection technology, since relays need on the order of 120 milliseconds to measure and breakers another 70 to open [6].

Then came 28 April 2025, and it was not a frequency event at all. Spain and Portugal went dark with load near 31 GW and solar supplying up to about 55 per cent of it. The precursors were oscillations — a local mode near 0.63 Hz traced to a photovoltaic plant in Badajoz, then a known inter-area mode near 0.21 Hz — on a 400 kV network being operated between 380 and 435 kV while generators were set to trip at 435 to 440 kV, a margin of a few volts per kilovolt. Roughly 2,180 MW disconnected in cascade over about two minutes as voltage climbed [14]. The expert panel convened by ENTSO-E, reporting in March 2026, attributed the blackout to interacting factors centred on oscillations, gaps in voltage and reactive-power control, divergent voltage-regulation practices and rapid generator disconnections in Spain [15]. Inertia does not appear among the causes.

## What the Metric Was Standing In For

There is a lesson in that sequence which is uncomfortable for the way the industry has organised its worrying. Inertia became the emblem of renewable-integration risk partly because it is genuinely fundamental and partly because it is a scalar. Megawatt-seconds can be metered, floored, tendered and audited; a control room can watch a single number fall through a threshold. Voltage stability and reactive-power margin are fields, distributed across thousands of nodes, and there is no equivalent number to put on a dashboard.

Meanwhile the thing being defended is quietly changing shape. A grid-forming inverter does not follow the network's frequency and phase; it imposes its own, and can be commanded to behave like a rotor of any chosen inertia constant while storing no kinetic energy whatsoever [10]. Its "seconds" are a control-loop parameter, tunable in firmware, bounded by converter headroom rather than by the tensile strength of forged steel. Whether that is a faithful substitute or a good imitation under the conditions tested so far is the open question, and the field data set is still thin — a handful of installations, a few years, no continental-scale disturbance yet ridden through by a majority-inverter system.

What is clear is that the seconds themselves were never the point. Inertia buys the interval in which something else — a governor valve, a battery, a relay that decides not to trip — has the chance to act. Grids have always been engineered around that interval. The interesting possibility is that a system built from converters could make it far shorter and not care, responding in milliseconds where a steam turbine needs seconds, so that the whole apparatus of inertia floors turns out to have been a proxy for response speed all along. The uncomfortable possibility is that the interval was doing quieter work too — holding voltage steady, supplying fault current, damping oscillations that nobody had to name because the spinning mass suppressed them for free. Iberia in 2025 is at least a hint that the second reading deserves a hearing.

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